Rational-Distance Points in the Unit Square

Research Archive: Algebraic Structures, Exact Searches,
Witness Elimination, and Geometry of the Four-Distance Surface

Yuan Si
University of Waterloo, Waterloo, Canada
y3si@uwaterloo.ca
ORCID: 0009-0005-1233-6972

April 2026

This archive documents a sustained investigation into whether any point in the interior of the unit square has all four vertex-distances rational. It contains four self-contained reports covering algebraic structures and structural obstructions, large-scale exact computational searches, the construction and elimination of an arithmetic witness, and the algebraic geometry of the four-distance surface.

The problem remains open. No claim of solution or non-solution is made. The search scripts and CAS verifications included in this archive use exact arithmetic and are independently reproducible.

Contents

Archive Overview **3**

1 Algebraic Structures and Obstructions **5**
1.1 Introduction . . . 5
1.2 Self-Incidence Coordinates . . . 5
1.3 The Curve Equation (†) . . . 6
1.4 The Genus-3 Curve C_{sw} . . . 6
1.5 The Intermediate Curve $\mathscr{C}_T$ . . . 6
1.6 The Elliptic Quotient E_x . . . 7
1.7 The Formula Chain (Steps 1–8) . . . 7
1.8 The Gröbner Basis Result . . . 7
1.9 The Quotient Ring Obstruction . . . 7
1.10 The Descent Circularity . . . 8
1.11 Counterexamples to Algebraic Sufficiency . . . 8
1.12 Summary . . . 8

2 Exact Computational Search **9**
2.1 Problem Statement . . . 9
2.2 Mod 3 Necessary Condition . . . 9
2.3 Two-Pythagorean-Triple Intersection Method . . . 10
2.4 Search Results . . . 10
2.5 Summary . . . 11
2.6 Conclusion . . . 11
2.7 Reproducibility . . . 11

3 Arithmetic Witnesses and Their Elimination **12**
3.1 Overview . . . 12
3.2 The Elliptic Curve $\mathscr{E}_B$ and the Function Ξ_B . . . 12
3.3 The Arithmetic Progression . . . 13
3.4 The $z = 14$ Witness . . . 13
3.5 Why Ξ_B Is Not Sufficient . . . 13
3.6 Exact Elimination of $z = 14$ . . . 14
3.7 Discussion . . . 14
3.8 Open Questions . . . 15

4 Geometry of the Four-Distance Surface **16**
4.1 Introduction 16
4.2 The Variety F_4 16
4.3 Dimension and Degree 17
4.4 Singularity Structure 17
4.5 Surface Type 18
4.6 Implications for the Rational-Distance Problem 18
4.7 Reproducibility 19
4.8 Summary 19

Archive Overview

Problem

For the unit square with vertices $A = (0,0)$, $B = (1,0)$, $C = (0,1)$, $D = (1,1)$: does there exist a point $P = (x,y)$ with $0 < y < x < 1$ such that all four distances PA, PB, PC, PD are rational?

This problem remains open.

What Is Established

The following results are rigorous within the scopes stated:

- A genus-3 hyperelliptic curve $C_{sw}\colon y^2 = w^8 - 6w^6 + 13w^4 - 8w^2 + 4$ arises naturally from the problem via self-incidence coordinates (Report 1).
- A chain of 8 algebraic maps connecting the geometric problem to C_{sw} is CAS-verified at each individual link, *except* the bridge from Stage-2 to $\mathscr{Q}_{-1}$, which is not verified (Report 1, §§ 7–8).
- Evidence that the natural algebraic reduction has a structural gap: the curve equation (†) is not a polynomial consequence of the distance-rationality conditions (Gröbner basis proof); explicit square-root extraction in the distance quotient ring fails (parity argument); and the 2-descent reformulation is equivalent to the gap itself (Report 1, §§ 8–10).
- No rational-distance point was found in an exact search covering all Pythagorean-compatible configurations with legs up to 5,000,000 (Report 2). This covers many but not all rational points with denominator up to $\sim$10,000,000.
- The $z = 14$ arithmetic witness, which appeared to demonstrate existence, is ruled out by an exact mod-193 obstruction on a specific discriminant. Only this final obstruction step is fully reproducible from this archive (Report 3).
- The four-distance variety $F_4 \subset \mathbb{A}^6$ is a surface of dimension 2, projective degree 16 in $\mathbb{P}^6$, with arithmetic genus 7 and 46 isolated singular points (vertex-collision configurations, verified as A_1 nodes at a representative point). The minimal resolution $\widetilde{F}_4$ is *not* K3 or rational; it is a surface of general type (or high-genus elliptic) (Report 4).

What Remains Open

- The existence or non-existence of an interior rational-distance point.
- Whether the 8 known rational points on C_{sw} are all rational points.
- Whether the Stage-2 to $\mathscr{Q}_{-1}$ bridge can be constructed.

Computational Artifacts

Verified scripts (used in final conclusions):

Script	Purpose	Deps
`direct_search.py`	Denominator sweep, $q \leq 100{,}000$	Python
`two_triple_search.py`	Two-triple join, legs $\leq$ 2M	Python
`search_5M.py`	Two-triple join, legs $\leq$ 5M	Python
`verify_identities.py`	Algebraic identities 1–6	SymPy
`verify_chain.py`	Formula chain Steps 1–8	SymPy
`verify_groebner.py`	Trivial elimination ideal	SymPy

Exploratory scripts (not used in final conclusions; see `exploratory/README_exploratory.md` in the scripts directory): Magma rank bound, Chabauty–Coleman analysis, p-adic logarithm, mod 3 local analysis.

Chapter 1

Algebraic Structures and Obstructions

1.1 Introduction

We study the question of whether there exists a point $P = (x, y)$ in the interior of the unit square with vertices $A = (0, 0)$, $B = (1, 0)$, $C = (0, 1)$, $D = (1, 1)$ such that PA, PB, PC, PD are all rational. We construct a genus-3 hyperelliptic curve C_{sw} associated to the problem, establish CAS-verified algebraic identities at individual chain links, and present evidence that the natural reduction strategy has a structural gap that is not closable within the algebraic frameworks we analyzed.

1.2 Self-Incidence Coordinates

Definition 1.2.1. For (x, y) with $0 < y < x < 1$, define

$$w := \frac{x+y}{x-y}, \qquad \sigma := \frac{2-x-y}{x-y}. \tag{1.1}$$

The inverse is $x = (w+1)/(w+\sigma)$, $y = (w-1)/(w+\sigma)$. For interior points of the basic domain: $w > 1$ and $\sigma > 0$.

Lemma 1.2.2 (CAS-verified; `verify_identities.py`)**.** *In (w, σ)-coordinates, the squared distances are*

$$PA^2 = \frac{2(w^2+1)}{(w+\sigma)^2}, \qquad PB^2 = \frac{(w-1)^2+(\sigma-1)^2}{(w+\sigma)^2}, \tag{1.2}$$

$$PC^2 = \frac{(w+1)^2+(\sigma+1)^2}{(w+\sigma)^2}, \qquad PD^2 = \frac{2(\sigma^2+1)}{(w+\sigma)^2}. \tag{1.3}$$

All four distances are rational if and only if

$$\text{(A):} \quad 2(1+w^2) \in \mathbb{Q}^{\times 2}, \tag{1.4}$$

$$\text{(D):} \quad 2(1+\sigma^2) \in \mathbb{Q}^{\times 2}, \tag{1.5}$$

$$\text{(B):} \quad (w-1)^2+(\sigma-1)^2 \in \mathbb{Q}^{\times 2}, \tag{1.6}$$

$$\text{(C):} \quad (w+1)^2+(\sigma+1)^2 \in \mathbb{Q}^{\times 2}. \tag{1.7}$$

1.3 The Curve Equation (†)

Definition 1.3.1. The curve equation (†) is

$$(3-w^2)\sigma^2 + (w^4-w^2-2)\sigma + w^2(1-w^2) = 0. \tag{†}$$

Proposition 1.3.2 (CAS-verified; `verify_identities.py`)**.** *Define* $y_E := 2(3-w^2)\sigma+(w^4-w^2-2)$. *Then in* $\mathbb{Q}[w,\sigma]$*:*

$$y_E^2 - P_4(w^2) = 4(3-w^2)\cdot(\dagger) \tag{1.8}$$

where $P_4(u) = u^4-6u^3+13u^2-8u+4$. *For* $w^2 \neq 3$*:* $(\dagger) = 0$ *if and only if* $y_E^2 = P_4(w^2)$.

1.4 The Genus-3 Curve C_{sw}

Definition 1.4.1. $C_{sw}\colon y^2 = w^8 - 6w^6 + 13w^4 - 8w^2 + 4 = P_4(w^2)$.

Properties:

- Genus 3. Smooth over $\mathbb{Q}$. Geometrically irreducible (CAS-verified: $P_4(w^2)$ is squarefree over $\overline{\mathbb{Q}}$).
- $\mathrm{disc}(P_4) = 2^{10}\cdot 41$.
- Known rational points: $\{(0,\pm 2),\ (\pm 1,\pm 2),\ \infty_\pm\}$ (8 points, all with $|w|\leq 1$).
- An exploratory Magma computation gives $1 \leq \mathrm{rank}(J(C_{sw})(\mathbb{Q})) \leq 2$ via 2-Selmer methods. This rank bound is not used in any final conclusion of this archive.

1.5 The Intermediate Curve $\mathscr{C}_T$

Definition 1.5.1. $\mathscr{C}_T\colon n^2 = 2t^4 + 13t^2 + 16$ (genus 1).

Connecting equations (Step 8):

$$w^2 = \frac{3t^2+tn+4}{2(t^2+2)}, \qquad \sigma = \frac{n+t}{n-t}. \tag{1.9}$$

Proposition 1.5.2 (CAS-verified; `verify_identities.py`, Identity 3)**.** *Invert the connecting substitution* (1.9) *to express* t^2 *and* n^2 *as rational functions of* (w,σ)*. Write* $F(w,\sigma)$ *for the numerator of* $n^2 - (2t^4+13t^2+16)$ *after clearing all denominators. Then*

$$\frac{F(w,\sigma)}{(\dagger)} = -8\sigma \tag{1.10}$$

as a rational function identity in $\mathbb{Q}(w,\sigma)$*. In particular,* $(t,n)\in\mathscr{C}_T(\mathbb{Q})$ *implies* $(\dagger) = 0$ *(when* $\sigma\neq 0$*).*

Clarification. The identity is verified in (w,σ)-coordinates, *not* as a polynomial identity in (t,n). The CAS (`verify_identities.py`, Identity 3) substitutes the inverse of (1.9), clears denominators, and checks that the ratio of the resulting numerator to (†) equals exactly -8σ. The original relation in (t,n)-coordinates is

$$(\dagger) = -\frac{t(n+t)}{(n-t)^2(t^2+2)^2}\left(n^2 - (2t^4+13t^2+16)\right),$$

a rational (non-polynomial) identity obtained by direct substitution of (1.9). Both forms give the same implication $\mathscr{C}_T(\mathbb{Q}) \Rightarrow (\dagger) = 0$.

1.6 The Elliptic Quotient E_x

The involution $\kappa\colon (w,y) \mapsto (-w,y)$ on C_{sw} gives a degree-2 quotient $\pi\colon C_{sw} \to \mathscr{E}_{sw}$ with $\mathscr{E}_{sw}\colon y^2 = P_4(u)$, $u = w^2$. In Weierstrass form:

$$E_x\colon Y^2 = X(X^2 + 13X + 32). \tag{1.11}$$

An exploratory Magma computation gives $\mathrm{rank}(E_x(\mathbb{Q})) = 1$, $E_x(\mathbb{Q})_{\mathrm{tors}} = \mathbb{Z}/2\mathbb{Z}$, generator $G = (-8, 8)$.

1.7 The Formula Chain (Steps 1–8)

Eight explicit rational maps form a chain:

$$(x,y,a,b,c,d) \to (r,s_0,\tau) \to (V,X,Y) \to (w,\sigma) \to (\rho,\omega) \in \mathscr{Q}_{-1} \to E_x \to \mathscr{E}_B \to \mathscr{C}_T \to (w^2,\sigma).$$

Each link *except* the bridge from Stage-2 to $\mathscr{Q}_{-1}$ is CAS-verified by polynomial division (`verify_chain.py`, `verify_identities.py`).

The gap. The map from the Stage-2 output (the four-distance locus) to the quartic 2-covering $\mathscr{Q}_{-1}\colon \omega^2 = -4\rho^4 + 13\rho^2 - 8$ is described but neither constructed explicitly nor CAS-verified. This is the principal unresolved step in the algebraic reduction.

1.8 The Gröbner Basis Result

Theorem 1.8.1 (CAS-verified; `verify_groebner.py`)**.** *Let $I = \langle f_1, f_2, f_3, f_4\rangle \subset \mathbb{Q}[w,\sigma,p,q,\lambda,\mu]$ encode conditions (1.4)–(1.7) via Pythagorean and Pell parametrizations. Then*

$$I \cap \mathbb{Q}[w,\sigma] = (0).$$

In particular, the curve equation (†) is not a polynomial consequence of the four distance-rationality conditions.

1.9 The Quotient Ring Obstruction

Proposition 1.9.1. *Let $R = \mathbb{Q}[x,y,a,b,c,d]/I_{\mathrm{dist}}$ where I_{dist} is generated by*

$$a^2 - x^2 - y^2, \quad b^2 - (1-x)^2 - y^2,$$
$$c^2 - x^2 - (1-y)^2, \quad d^2 - (1-x)^2 - (1-y)^2.$$

The descent numerator $N(x,y)/(4x^2y^2)$ from the birational map $\mathscr{E}_{sw} \to E_x$ does not appear to be a square in $\mathrm{Frac}(R)$.

Evidence. $N(x,y) = (x-y)(x^3-5x^2y-2x^2-xy^2+8xy+y^3-2y^2)$ contains the factor $(x-y)$ to odd multiplicity (the cofactor evaluates to $-4x^2(x-1) \neq 0$ at $y = x$). For each of the 15 nontrivial squarefree distance monomials $M \in \{a,b,c,d,ab,\ldots,abcd\}$, $M^2|_{y=x} \neq 0$ (verified by evaluation). Under the assumption that the monomials form a free basis for R over $\mathbb{Q}[x,y]$—which holds generically but for which we have not included a complete field-theoretic proof—$N/(4x^2y^2)$ is not a square in $\mathrm{Frac}(R)$.

Remark. A complete proof would require verifying that R is an integral domain and that the monomials are indeed $\mathbb{Q}[x,y]$-linearly independent. We present this as strong evidence rather than a fully rigorous impossibility result.

1.10 The Descent Circularity

Proposition 1.10.1 (CAS-verified)**.** *The cubic in $\alpha = s^2$ arising from the w^2 connecting equation factors as*

$$-2(\alpha+2)\big(\alpha^2 - (2w^4{-}6w^2)\alpha - (4w^4{-}8w^2{+}4)\big) = 0.$$

The residual quadratic has discriminant

$$(2w^4{-}6w^2)^2 + 4(4w^4{-}8w^2{+}4) = 4P_4(w^2).$$

For $w^2 \neq 3$, this discriminant is a rational square if and only if $P_4(w^2)$ is a rational square, if and only if $(\dagger) = 0$. (At $w^2 = 3$: $P_4(3) = 16 = 4^2$, so $4P_4(3) = 64$ is a square regardless of $(\dagger)$; but $w^2 = 3$ is irrational and does not arise from rational geometric data.) Hence the descent into $\mathscr{C}_T$ is logically equivalent to $(\dagger)$ for all rational w: the 2-descent on E_x does not provide an independent route, within the framework analyzed here.

1.11 Counterexamples to Algebraic Sufficiency

Observation 1.11.1. There exist $(w, \sigma) \in \mathbb{Q}^2$ satisfying all four conditions (1.4)–(1.7) simultaneously with $(\dagger) \neq 0$.

Example: $(w, \sigma) = (7, -7)$.

- $2(1{+}49) = 100 = 10^2$. Condition (1.4) holds.
- $2(1{+}49) = 100 = 10^2$. Condition (1.5) holds.
- $6^2 + (-8)^2 = 100 = 10^2$. Condition (1.6) holds.
- $8^2 + (-6)^2 = 100 = 10^2$. Condition (1.7) holds.
- $(\dagger) = (3{-}49)(49) + (2401{-}49{-}2)(-7) + 49(1{-}49) = -21056 \neq 0$.

This has $\sigma < 0$ (not a geometric interior point), but shows that $(\dagger)$ is not a formal consequence of (A)–(D) over $\mathbb{Q}$.

1.12 Summary

Claim	**Status**	**Verification**
Distance formulas	CAS-verified	`verify_identities.py`
$y_E^2 - P_4 = 4(3{-}w^2)(\dagger)$	CAS-verified	`verify_identities.py`
Chain links (7 of 8, excl. bridge)	CAS-verified	`verify_chain.py`
Bridge: Stage-2 to $\mathscr{Q}_{-1}$	**Not constructed**	Gap
$I \cap \mathbb{Q}[w, \sigma] = (0)$	CAS-verified	`verify_groebner.py`
No ρ in $\mathrm{Frac}(R)$	Evidence (Prop. 1.9.1)	Parity argument
Descent disc. $= 4P_4(w^2)$	CAS-verified	CAS factorization
Descent $\Leftrightarrow$ $(\dagger)$	Proved (within framework)	Algebraic argument

The intended reduction "geometric solution $\Rightarrow$ C_{sw} point with $w > 1$" has an unverified step. Within the frameworks analyzed, attempts to close this gap—via quotient-ring methods (Proposition 1.9.1) or 2-descent (Proposition 1.10.1)—do not succeed.

Chapter 2

Exact Computational Search

2.1 Problem Statement

We search for rational (x, y) with $0 < y < x < 1$ such that all four vertex-distances of the unit square are rational. Writing $x = p/q$, $y = r/q$ with integers $0 < r < p < q$ and $\gcd(p, r, q) = 1$, the conditions become: the four integers

$$p^2+r^2, \quad (q-p)^2+r^2, \quad p^2+(q-r)^2, \quad (q-p)^2+(q-r)^2$$

must all be perfect squares. No floating-point arithmetic is used.

2.2 Mod 3 Necessary Condition

Theorem 2.2.1. *In any solution with* $\gcd(p, r, q) = 1$*:* $3 \mid q$*, and exactly one of* p, r *is divisible by* 3*.*

Proof. Quadratic residues mod 3 are $\{0, 1\}$; the value 2 is a non-residue. The sum $a^2 + b^2 \equiv 2 \pmod 3$ iff both $a, b \not\equiv 0 \pmod 3$, so such a sum cannot be a perfect square.

Case $3 \nmid q$*.* From condition (I): $3 \mid p$ or $3 \mid r$. If $3 \mid p$: then $3 \nmid (q-p)$, so from (II): $3 \mid r$. But then $3 \mid \gcd(p, r)$ with $3 \nmid q$: contradicts $\gcd(p, r, q) = 1$. If $3 \nmid p$: then $3 \mid r$ (from I). From (III): $3 \mid p$ or $3 \mid (q-r)$; since $3 \nmid p$ and $3 \nmid q$ but $3 \mid r$: $3 \nmid (q-r)$, so $3 \mid p$. Contradiction.

Case $3 \mid q$*.* $(q-p) \equiv -p \pmod 3$ and $(q-r) \equiv -r \pmod 3$. All four conditions reduce to $p^2+r^2 \not\equiv 2 \pmod 3$, i.e., $3 \mid p$ or $3 \mid r$. If both hold: $3 \mid \gcd(p, r, q)$, contradicting primitivity. So exactly one of p, r is divisible by 3. □

Remark. This is a necessary condition on solutions in the integer model $x = p/q$, $y = r/q$. It does *not* constitute a local obstruction at $p = 3$ for the original problem over $\mathbb{Q}$, because $\mathbb{Q}_3$-solutions exist with 3 in the denominator of x or y.

2.3 Two-Pythagorean-Triple Intersection Method

Setting $s = q - p$ and $t = q - r$, the four conditions become Pythagorean membership tests:

$$p^2 + r^2 = a^2 \iff p \in \text{partners}(r), \tag{2.1}$$
$$s^2 + r^2 = b^2 \iff s \in \text{partners}(r), \tag{2.2}$$
$$p^2 + t^2 = c^2 \iff t \in \text{partners}(p), \tag{2.3}$$
$$s^2 + t^2 = d^2 \iff t \in \text{partners}(s), \tag{2.4}$$

with $q = p + s$ and $t = p + s - r$, where $\text{partners}(L) = \{M > 0 : L^2 + M^2 \text{ is a perfect square}\}$.

Algorithm.

1. **Index construction.** For each $L \leq$ `LEG_MAX`, compute partners(L) using the (m, n, k)-parametrization of Pythagorean triples.
2. **Common-leg iteration.** For each common leg r, retrieve partners(r). For each pair (p, s) from partners(r) with $p > r$, $s > 0$: set $q = p+s$, $t = q-r$; apply mod 3 filter and coprimality check; test $t \in$ partners(p) and $t \in$ partners(s) (each an $O(1)$ set lookup).

Coverage note. The search covers all Pythagorean-compatible configurations (p, r, s, t) with all four legs $\leq$ `LEG_MAX`. This includes many rational points with denominator $q = p+s$ up to $2\cdot$`LEG_MAX`, but is *not* equivalent to covering all denominators up to that bound: some configurations with large legs and moderate denominator may be missed.

2.4 Search Results

Run 1: Denominator sweep

Script: `direct_search.py` with $q \leq 100{,}000$.

Range	$q \leq 100{,}000$ (multiples of 3 only)
Combinations tested	1,476,656,036
Passed first square test	10,053
Solutions	0
Wall time	$\sim$606 s

Run 1 is a true denominator sweep: it covers *all* $q \leq 100{,}000$ with $3 \mid q$.

Run 2: Two-triple intersection, `LEG_MAX` = 2 M

Script: `two_triple_search.py`.

Legs indexed	$\sim$1,000,000
Pairs tested	184,778,466
Condition-3 survivors	196
Condition-4 survivors	0
Max leg	2,000,000
Wall time	54 s

Run 3: Two-triple intersection, `LEG_MAX` = 5 M

Script: `search_5M.py`.

Legs indexed	3,409,677
Partner entries	25,503,012
Pairs tested	585,651,073
Condition-3 survivors	297
Condition-4 survivors	0
Max leg	5,000,000
Wall time	183 s

2.5 Summary

Metric	**Run 1**	**Run 2**	**Run 3**
Method	Denom. sweep	Two-triple	Two-triple
Coverage	all $q \leq 10^5$	legs $\leq$ 2M	legs $\leq$ 5M
Combinations	1.48 B	185 M	586 M
First-square pass / cond-3	10,053	196	297
Solutions	0	0	**0**
Wall time	606 s	54 s	183 s

2.6 Conclusion

No rational-distance point was found. Run 1 provides an exact denominator guarantee: no solution exists with $q \leq 100{,}000$. Runs 2–3 provide Pythagorean-leg guarantees: no solution exists whose four distance numerators involve legs up to 5,000,000. The latter covers a much larger effective range but is not a complete denominator sweep beyond $q = 100{,}000$.

2.7 Reproducibility

All scripts use only the Python standard library. To reproduce Run 3:

```
python search_5M.py
```

Expected output: `NO GENUINE RATIONAL-DISTANCE POINT FOUND` after approximately 3 minutes.

Chapter 3

Arithmetic Witnesses and Their Elimination

3.1 Overview

During the investigation, a candidate arithmetic witness ($z = 14$) was identified that satisfies a first-layer necessary condition for the existence of a rational-distance point. This report documents its construction, the condition it satisfies, the additional condition it fails, and the exact finite-field obstruction that eliminates it.

The derivation of the witness quantity Ξ_B and its relationship to the original geometric problem involves a multi-step reduction chain documented in external phase files. Only the final obstruction step (the mod-193 elimination in § 6) is fully self-contained and reproducible from this archive alone.

3.2 The Elliptic Curve $\mathscr{E}_B$ and the Function Ξ_B

Definition 3.2.1. The elliptic curve

$$\mathscr{E}_B \colon V^2 = U(U^2 - 26U + 41) \tag{3.1}$$

has a rational point $P = (25, 20)$ of infinite order.

Definition 3.2.2. For (U, V) on $\mathscr{E}_B$ with $Q_4(U) \neq 0$:

$$\Xi_B(U, V) = \frac{R_4(U) - 4(U^2 - 41)V}{Q_4(U)}, \tag{3.2}$$

where

$$Q_4(U) = U^4 - 20U^3 - 330U^2 - 820U + 1681, \tag{3.3}$$

$$R_4(U) = U^4 - 28U^3 - 378U^2 - 1148U + 1681. \tag{3.4}$$

The function Ξ_B arose as the composition $\Xi_B = \xi \circ \Psi$, where ξ is a descent map on E_x and $\Psi \colon \mathscr{E}_B \to E_x$ is an isogeny. In the intended reduction chain, the condition $\Xi_B(Q) \in \mathbb{Q}^{\times 2} \cap (9/5,\, 2)$ was identified as a necessary condition for the existence of an interior rational-distance point in the corresponding arithmetic family.

3.3 The Arithmetic Progression

For each integer z, define

$$Q_z := [3402000 \cdot z]\, P \;\in\; \mathscr{E}_B(\mathbb{Q}). \tag{3.5}$$

Local squareness (from phases P241/P242). The phases argue that for every $z \in \mathbb{Z}$ and every finite prime p: $\Xi_B(Q_z) \in \mathbb{Q}_p^{\times 2}$. This is attributed to the 2-descent structure: $\Psi(Q_z)$ lies in $B + 2E_x(\mathbb{Q})$. If correct, the Hasse–Minkowski theorem implies: $\Xi_B(Q_z) > 0 \Rightarrow \Xi_B(Q_z) \in \mathbb{Q}^{\times 2}$.

We note that this local squareness claim is taken from the phase files and has not been independently CAS-verified in this archive.

3.4 The $z = 14$ Witness

$Q_{14} = [47628000]P$ on $\mathscr{E}_B$.

Certified bounds (phase P245, outward-rounded interval arithmetic):

$$U_{14} \in [24.40657345591636632758,\; 24.40657345591636632759], \tag{3.6}$$

$$V_{14} \in [-7.17606220571288309434,\; -7.17606220571288309434]. \tag{3.7}$$

With the admissible window $\rho = 13{+}8\sqrt{2} < U_{14} < U_{9/5}$ and $V_{14} < 0$, and assuming local squareness:

$$\Xi_B(Q_{14}) \in \mathbb{Q}^{\times 2} \cap \left(\tfrac{9}{5},\, 2\right). \tag{3.8}$$

This was initially interpreted (in phase P245) as proving the existence of an interior rational-distance point. That interpretation was premature, as shown below.

3.5 Why Ξ_B Is Not Sufficient

The condition $\Xi_B(Q) \in \mathbb{Q}^{\times 2} \cap (9/5, 2)$ is necessary but not sufficient. The full recovery requires three layers:

Layer 1. Self-incidence parameter. $r^2 = \Xi_B(Q)/(2{-}\Xi_B(Q))$. Rational whenever $\Xi_B(Q)$ is rational. Automatic.

Layer 2. Companion parameter. $s = (\eta - 2\rho_B)/(\eta + 2\rho_B)$ where $\rho_B = V/U$ and $\eta = U - 41/U$. Always a well-defined rational number (for $U \neq 0$ and $\eta + 2\rho_B \neq 0$).

Layer 3. Side-distance rationality. The variable $u_\natural$, defined by the quadratic

$$u^2 - 4(r^2{+}s^2{+}2)\, u + 16(r{+}s)^2 = 0,$$

must be rational. This requires the discriminant

$$\delta_u = (r^2{+}s^2{+}2)^2 - 4(r{+}s)^2 \tag{3.9}$$

to be a rational perfect square. This condition is *not* guaranteed by the previous layers.

3.6 Exact Elimination of $z = 14$

Method. Test Layer 3 modulo a prime of good reduction, using only finite-field arithmetic.

Prime. $p = 193$. Good reduction: $\mathrm{disc}(\mathscr{E}_B) = 2^{13} \cdot 41^2$, and 193 divides neither 2 nor 41.

Step 1. Exact double-and-add on $\mathscr{E}_B(\mathbb{F}_{193})$:

$$Q_{14} \equiv (112,\, 158) \pmod{193}.$$

Step 2. Derived quantities mod 193:

$$\Xi_B(Q_{14}) \equiv 115, \qquad r^2 \equiv 110, \tag{3.10}$$

$$\rho_B \equiv 60, \qquad \eta \equiv 72, \tag{3.11}$$

$$s \equiv 48. \tag{3.12}$$

Step 3. $r^2 \equiv 110 \pmod{193}$ has two square roots: $r \equiv 66$ or $r \equiv 127$. Evaluate δ_u for each (all arithmetic mod 193):

$$\begin{aligned} r \equiv 66\colon &\quad \delta_u \equiv 100^2 - 4 \cdot 114^2 \equiv 90, \\ r \equiv 127\colon &\quad \delta_u \equiv 100^2 - 4 \cdot 175^2 \equiv 19. \end{aligned}$$

Step 4. Euler criterion:

$$\begin{aligned} 90^{96} &\equiv -1 \pmod{193} \quad \text{(non-residue)}, \\ 19^{96} &\equiv -1 \pmod{193} \quad \text{(non-residue)}. \end{aligned}$$

Both are quadratic non-residues.

Proposition 3.6.1. *If the formulas in Steps 1–2 correctly reduce the global quantities $\Xi_B(Q_{14})$, r^2, s to $\mathbb{F}_{193}$, then δ_u is not a square in $\mathbb{F}_{193}$ for either choice of r. Since a rational perfect square reduces to a square modulo every prime of good reduction, $\delta_u \notin \mathbb{Q}^{\times 2}$. Therefore $u_{\mathfrak{q},14} \notin \mathbb{Q}$, and the $z = 14$ witness does not correspond to a genuine geometric solution.*

Remark. The finite-field arithmetic in Steps 1–4 is self-contained and verifiable. The formulas connecting Ξ_B, r^2, s to the original geometric problem are taken from the phase-file derivation and have not been independently re-derived in this archive.

3.7 Discussion

1. **Necessary vs. sufficient.** Ξ_B being a rational square is a first-layer condition. Passing one rationality layer does not guarantee passing the next.
2. **Efficiency of modular elimination.** A single prime ($p = 193$) sufficed. The computation uses only finite-field arithmetic with numbers below 193.
3. **Status of the P245 claim.** The assertion $\mathcal{S} \neq \varnothing$ in phase P245 was based solely on Layer 1. With Layer 3 failing at $p = 193$, the assertion is retracted.

3.8 Open Questions

1. Are there other values of z for which all three layers are satisfied? The modular test can be applied systematically.
2. Can the screening be extended to a global proof that no z works? This would require showing δ_u is never a rational square on the entire progression—a problem of independent arithmetic interest.

Chapter 4

Geometry of the Four-Distance Surface

4.1 Introduction

This report records the computational investigation of the algebraic geometry of the four-distance variety F_4, carried out with Magma. The goal was not to prove non-existence of rational-distance points but to understand the geometric type of the variety. Knowing whether F_4 is rational, K3, elliptic, or general type determines which tools (if any) are applicable for future Brauer-Manin or Mordell-Weil analysis.

All dimensions, degrees, and singularity data reported here were computed by Magma and can be reproduced with the scripts listed in Section 4.7.

4.2 The Variety F_4

Definition 4.2.1. $F_4 \subset \mathbb{A}^6_{\mathbb{Q}}$ is the affine variety defined by the four quadric equations

$$f_1 := a^2 - x^2 - y^2 = 0, \tag{4.1}$$
$$f_2 := b^2 - (1-x)^2 - y^2 = 0, \tag{4.2}$$
$$f_3 := c^2 - x^2 - (1-y)^2 = 0, \tag{4.3}$$
$$f_4 := d^2 - (1-x)^2 - (1-y)^2 = 0. \tag{4.4}$$

A point $(x, y, a, b, c, d) \in F_4(\mathbb{Q})$ with $0 < y < x < 1$ and $a, b, c, d > 0$ corresponds to a rational-distance point in the unit square.

Proposition 4.2.2 (British flag identity; CAS-verified)**.** *In the coordinate ring of F_4,*

$$a^2 + d^2 = b^2 + c^2. \tag{4.5}$$

Equivalently, $a^2 - b^2 - c^2 + d^2 \in (f_1, f_2, f_3, f_4)$.

Proof. $f_1 - f_2 - f_3 + f_4 = (a^2 - b^2 - c^2 + d^2) + R(x, y)$ for some polynomial R in x, y that vanishes identically. Direct expansion: $R(x, y) = 0$. Verified by Magma's `NormalForm`. □

4.3 Dimension and Degree

Theorem 4.3.1 (Magma-verified)**.** *F_4 is a reduced, irreducible affine surface of dimension* 2*.*

Explicit data (Magma):
- $\dim F_4 = 2$.
- F_4 is reduced: $I(F_4) = \sqrt{I(F_4)}$.
- F_4 is irreducible as a scheme over $\mathbb{Q}$.

Projective closure

Homogenizing each f_i with respect to a new variable W yields the homogeneous system

$$F_1 = A^2 - X^2 - Y^2, \tag{4.6}$$
$$F_2 = B^2 - (W - X)^2 - Y^2, \tag{4.7}$$
$$F_3 = C^2 - X^2 - (W - Y)^2, \tag{4.8}$$
$$F_4 = D^2 - (W - X)^2 - (W - Y)^2, \tag{4.9}$$

and a projective surface $\overline{F_4} \subset \mathbb{P}^6$ with coordinates $[X : Y : A : B : C : D : W]$.

Theorem 4.3.2 (Magma-verified)**.** *$\overline{F_4} \subset \mathbb{P}^6$ is a reduced, irreducible projective surface of degree* 16*. Its Hilbert polynomial is*

$$P(n) = 8n^2 - 8n + 8. \tag{4.10}$$

Corollary 4.3.3. *The arithmetic genus of $\overline{F_4}$ is*

$$p_a(\overline{F_4}) = P(0) - 1 = 7. \tag{4.11}$$

4.4 Singularity Structure

Theorem 4.4.1 (Magma-verified)**.** *$\overline{F_4}$ has a* 0*-dimensional singular subscheme of degree* 46 *(counted with scheme-theoretic multiplicity). Specifically:*
- 32 *singular points in the affine part F_4,*
- 12 *singular points at infinity $W = 0$ (the remaining* 2 *come from higher-order contributions to the singular scheme structure).*

There are no $\mathbb{Q}$-rational singular points. All singularities live over quadratic extensions of $\mathbb{Q}$, typically $\mathbb{Q}(\sqrt{2})$.

Geometric description of the affine singularities

Magma's Gröbner basis of the singular ideal reveals that every affine singular point satisfies:

$$a \cdot b \cdot c \cdot d = 0.$$

Combined with the remaining defining equations, this means at least one of the four distance coordinates vanishes.

Proposition 4.4.2. *The affine singular points of F_4 correspond to the four "vertex-collision" configurations:*
- $a = 0 \iff (x, y) = (0, 0)$ *(P at vertex A)*

- $b = 0 \iff (x, y) = (1, 0)$ *(P at vertex B)*
- $c = 0 \iff (x, y) = (0, 1)$ *(P at vertex C)*
- $d = 0 \iff (x, y) = (1, 1)$ *(P at vertex D)*

Each vertex gives 8 *singular points via sign choices in the remaining three distance coordinates, for a total of* 32.

Singularity type

Proposition 4.4.3 (Magma-verified at one point)**.** *The singular point* $(x, y, a, b, c, d) = (0, 0, 0, 1, 1, \sqrt{2})$ *is an ordinary double point* (A_1 *node*) *of* F_4.

Verification. The Jacobian matrix of (f_1, f_2, f_3, f_4) evaluated at this point has rank 3 over $\mathbb{Q}(\sqrt{2})$, strictly less than the codimension $\operatorname{codim}(F_4 \subset \mathbb{A}^6) = 4$, confirming singularity. The Hessian of f_1 restricted to the smooth 3-fold locally cut out by $\{f_2 = f_3 = g = 0\}$ (where $g = a^2 - b^2 - c^2 + d^2$) has rank 3, which is the condition for an A_1 node in this codim. □

By symmetry (rotations and reflections of the square permute the four vertex configurations), the analogous statement holds at each of the 32 affine singular points, giving:

Observation 4.4.4. All 32 affine singular points of F_4 are A_1 nodes. This has only been verified computationally at one representative point; the remaining cases follow by Magma symmetry arguments but have not been independently re-verified point by point.

4.5 Surface Type

Theorem 4.5.1 (From Corollary 4.3.3)**.** *Let* $\widetilde{F}_4$ *denote the minimal resolution of* $\overline{F_4}$. *If all* 46 *singularities of* $\overline{F_4}$ *are* A_1 *nodes, then:*

1. $\widetilde{F}_4$ *is smooth.*
2. *The geometric genus of* $\widetilde{F}_4$ *satisfies* $p_g(\widetilde{F}_4) = p_a(\overline{F_4}) = 7$.
3. $\widetilde{F}_4$ *is* not *a K3 surface* (*which would require* $p_g = 1$).
4. $\widetilde{F}_4$ *is* not *a rational or birationally ruled surface* (*which would require* $p_g = 0$).

Corollary 4.5.2. $\widetilde{F}_4$ *is a* surface of general type *or an elliptic surface of positive geometric genus.*

4.6 Implications for the Rational-Distance Problem

Theorem 4.5.1 has two consequences for attacks on the rational-distance problem:

1. **Faltings' theorem does not apply directly.** Faltings' theorem gives finiteness of rational points on curves of genus ≥ 2, not surfaces. Direct application to $\widetilde{F}_4$ is not available.

2. **Brauer-Manin obstruction theory is substantially harder for surfaces of general type than for K3 surfaces or rational surfaces.** For K3 surfaces, the Brauer group is well-understood and there are algorithmic approaches. For surfaces of general type, Brauer-Manin theory is mathematically developed but computational algorithms are not standardized. A Brauer-Manin analysis of $\widetilde{F}_4$ would be a research project, not a routine computation.

3. **Elliptic-fibration approaches may be viable.** $\widetilde{F}_4$ admits natural fibrations in each variable. Analyzing rational points fiber by fiber (using Mordell finiteness on fibers of genus ≥ 2) is a direction that does not appear to have been tried in the literature.

4.7 Reproducibility

All computations in this report were performed using Magma (online calculator at magma.maths.usyd.edu.au/calc). The scripts are included in the archive under `scripts/geometry/`:

- `f4_geometry_fixed.magma`: dimension, degree, radical check, singular locus dimension.
- `f4_singularities.magma`: boundary at infinity, Hilbert polynomial, singular subscheme at infinity.
- `f4_node_check.magma`: Gröbner basis of the singular ideal revealing vertex-collision structure.
- `f4_nodality.magma`: verification of A_1 nodality at one representative point over $\mathbb{Q}(\sqrt{2})$.

Each script takes well under 120 seconds on the online calculator.

4.8 Summary

Property	**Value / Status**
Dimension of F_4	2 (surface)
Irreducible, reduced	yes
Projective degree in $\mathbb{P}^6$	16
Hilbert polynomial	$8n^2 - 8n + 8$
Arithmetic genus p_a	7
Singular subscheme degree	46
Rational singular points	0
Singularity type at $(0, 0, 0, 1, 1, \sqrt{2})$	A_1 node
All 32 affine sing. points	A_1 by symmetry (not independently verified)
Geometric genus $p_g(\widetilde{F}_4)$	7 (if all nodes)
Surface type of $\widetilde{F}_4$	General type or high-genus elliptic
K3?	No
Rational/ruled?	No